

Anubhav Chakraborty, PhD

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Professional Summary

Molecular biologist and genetic engineer with a PhD in Molecular and Integrative Physiology (awarded August 2026) and 7 years of research experience specializing in functional genomics, *in vitro* kidney models, target discovery, CRISPR-based genome engineering, epigenetic modulation and viral/non-viral gene delivery platforms for kidney diseases. Proven track record in primary cell culture (human & mouse renal epithelial cells), cell line development, cloning, viral (AAV, lentivirus) and non-viral (LNPs, VLPs) vector engineering. AHA Predoctoral Fellow and back-to-back ASN Kidney STARS awardee with a pending US patent for CRISPR-based gene therapy. Independent researcher who is skilled in complex data analysis, presenting at national conferences, mentoring junior researchers, and collaborating across multidisciplinary teams.

Education/Employment

University of Kansas Medical Center

Doctor of Philosophy (PhD) in Molecular and Integrative Physiology

Cumulative GPA: 3.88/4.0

Kansas City, KS, USA

Aug 2019 – Aug 2026

Vellore Institute of Technology

Council of Scientific and Industrial Research (CSIR) - Govt. of India,

Senior Research Fellow

Tamil Nadu, India

Dec 2017 – Jun 2019

SRM Institute of Science and Technology

Master of Technology (MTech) in Genetic Engineering

First Class (CGPA: 7.23/10)

Tamil Nadu, India

Jun 2015 – May 2017

SASTRA University

Bachelor of Technology (BTech) in Bioengineering

First Class (CGPA: 7.33/10)

Tamil Nadu, India

Jun 2011 – May 2015

Technical Skills & Scientific Capabilities

- **Vector Engineering & Gene Delivery:** Recombinant AAV (rAAV) and lentivirus design and production, Virus-Like Particles (VLPs), Lipid Nanoparticles (LNPs), Nucleofection, Mini-circle production.
- **Molecular Biology:** CRISPR activation (CRISPRa) and interference (CRISPRi), CRISPR-Cas9/Cas12a gene editing, Chromatin accessibility assays, Protein trans-splicing systems, Advanced molecular cloning, ELISA, Immunoblotting, Immunocytochemistry, RT-qPCR.
- ***In vitro* & *In Vivo* Disease Models:** Primary mouse renal epithelial cell isolation, Primary cell culture (mouse and human), 3D cell culture, Cell line development for disease modeling. Mouse surgical model of ischemia-reperfusion injury, Tissue micro-dissection.
- **Digital Tools & Lab Software:** Benchling & SnapGene (In silico vector design), LabArchive (Electronic Lab Notebook), GraphPad Prism, Microsoft Office Suite.

Research Experiences

The Jared Grantham Kidney Institute, University of Kansas Medical Center

Kansas City, KS, USA

Predoctoral candidate | Lab of Dr. Alan S. Yu

Aug 2020 – Present

- **CRISPRa Therapeutics:** Developed a CRISPRa-based gene therapy strategy (US patent pending) augmenting endogenous *PKD1* expression to rescue disease phenotypes in Autosomal Dominant Polycystic Kidney Disease (ADPKD).
- **Plasmid Engineering:** Engineered an innovative plasmid miniaturization process that significantly improved transgene delivery (by ~3x) and expression in hard-to-transfect primary renal epithelial cells.
- **Gene Delivery Platforms:** Designed, evaluated, and optimized diverse vector delivery systems including rAAVs, VLPs, Lentiviruses, Nucleofection and LNPs; for delivering complex genetic payloads across primary and immortalized cell lines.
- **Protein & Epigenetic Machinery:** Designed and investigated split-intein protein trans-splicing mechanisms to overcome packaging size constraints; conducted chromatin accessibility profiling to analyze heterochromatin limitations at the proximal promoter of several genes, benchmarked various CRISPR-effectors for optimal epigenomic modulation.
- **Consulting & Mentorship:** Served as a subject-matter expert advising inter-departmental laboratories on bespoke viral vector design, plasmid engineering, and stable line derivation; directly trained and mentored 4 junior researchers (1 intern, 3 pre-med students) in experimental design and molecular workflows.

Vellore Institute of Technology (VIT)

Tamil Nadu, India

Senior Research Fellow (CSIR) | Lab of Dr. Pragasam V

Dec 2017 – Jun 2019

- **Epigenomic Dynamics:** Investigated DNA methylation and demethylation dynamics during necroptotic cell death to elucidate epigenetic regulation during renal ischemia-reperfusion injury.
- **In vivo AKI Modeling:** Established surgical mouse models of ischemic acute kidney injury to analyze disease progression and cell death pathways.
- **Grant Proposals & Supervision:** Assisted faculty in technical drafting of competitive grant proposals; co-directed master's thesis execution for 4 graduate students.

SRM Institute of Science and Technology

Tamil Nadu, India

Master's student | Lab of Dr. J. Megala, Human Genetics Lab

Jan 2016 – May 2017

- **Genomic Polymorphisms:** Executed genotyping across 1,100 clinical subjects of Indian Tamil origin to determine genetic associations between TNF-alpha upstream promoter single-nucleotide polymorphisms and gastric mucosal inflammation caused by *Helicobacter pylori* infections.
- **Transcriptional Mechanics:** Established phylogenetic footprints and transcriptional regulation profiles of mutated TNF-alpha variants driving chronic inflammatory responses.

SASTRA University

Tamil Nadu, India

Bachelor's student | Lab of Dr. C. David Raj

Jul 2014 – May 2015

- **In Silico & In Vitro Assays:** Screened phytochemical lead molecules from extracts of *Cinnamomum tamala* and *Cinnamomum verum* leaf essential oils, performed molecular docking, and *ex vivo* superfusion assays on isolated rat ileum to analyze spasmolytic potential of eugenol.

Patent

Chakraborty, A., Wallace, D.P., and Yu, A.S.L. "CRISPRa Compositions, Methods of Use Thereof for Treating Polycystic Kidney Disease, and Methods of Making the Same." U.S. Provisional Patent Application No. 64/034,047, filed April 9, 2026.

Publications and Preprints

- **Chakraborty, A.**, Varghese, M.M, Wallace, D.P., Ward, C.J., Yu, A.S.L., 2026. CRISPR activation of endogenous PKD1 increases polycystin-1 levels and suppresses cellular features of ADPKD. bioRxiv. <https://doi.org/10.64898/2026.05.23.727418>
- **Chakraborty, A.** and Yu, A.S.L., 2026. Miniaturization of CRISPRa plasmids for efficient delivery into renal epithelial cells and Pkd1 transactivation. Molecular Biology Reports, 53 (707), <https://doi.org/10.1007/s11033-026-11872-1>
- Nyimanu, D., **Chakraborty, A.**, Parnell, S., Wallace, D. and Yu, A., 2026. Apelin inhibits cyst growth and improves kidney function in mice with polycystic kidney disease. bioRxiv, pp.2026-03. <https://doi.org/10.64898/2026.03.26.714294>
- Thornton, M., Sommer, N., McGonigle, M., Kumar Ram, A., Yerrathota, S., Ehirim, H., Chaturvedi, A., Dinh Phan, J., **Chakraborty, A.**, ...Sharma, M., 2025. Notch3 deletion regulates HIV-1 gene expression and systemic inflammation to ameliorate chronic kidney disease. Disease models & mechanisms, 18(2), p.DMM052056. <https://doi.org/10.1242/dmm.052056>
- **Chakraborty, A.** and Yu, A.S.L., 2025. Prospects for gene therapy in polycystic kidney disease. Current Opinion in Nephrology and Hypertension, 34(1), pp.121-127. <https://doi.org/10.1097/mnh.0000000000001030>
- **Chakraborty, A.**, Sankaran, V., Murugan, R. and Chellappan, D.R., 2021. Comparative spasmolytic effect between Cinnamomum tamala and Cinnamomum verum leaf essential oils and eugenol through in vitro and in silico approaches. Zeitschrift für Naturforschung C, 76(9-10), pp.383-391. <https://doi.org/10.1515/znc-2020-0153>
- **Chakraborty, A.** and Viswanathan, P., 2018. Methylation-demethylation Dynamics: Implications of Changes in Acute Kidney Injury. Analytical Cellular Pathology, 2018(1), p.8764384. <https://doi.org/10.1155/2018/8764384>
- **Chakraborty, A.**, Sankaran, V., Ramar, M. and Chellappan, D.R., 2015. Chemical analysis of leaf essential oil of Cinnamomum verum from Palni hills, Tamil Nadu. Tic, 3(e10), p.x10.
- Sankaran, V., **Chakraborty, A.**, Jeyaprakash, K., Ramar, M. and Chellappan, D.R., 2015. Chemical analysis of leaf essential oil of Cinnamomum tamala from Arunachal Pradesh, India. Journal of Chemical and Pharmaceutical Sciences, 8(2), pp.246-248.

Fellowships, Honors & Awards

- **American Heart Association (AHA) Predoctoral Fellowship:** Awarded \$71,828 national fellowship grant (Jan 2024 – Dec 2025). <https://doi.org/10.58275/AHA.24PRE1194472.pc.gr.190750>
- **ASN Kidney STARS Travel Award:** American Society of Nephrology (ASN) Kidney Week, Houston, TX (2025) & Philadelphia, PA (2023).
- **ASN Kidney TREKS Scholar:** Selected for MDBIL Research Program, Bar Harbor, ME (2024).
- **Biomedical Research Training Program Scholarship:** Awarded \$13,000 – University of Kansas Medical Center (2023).
- **KUMC Physiology Society Travel Award:** University of Kansas Medical Center (2023).
- **Best Poster Award:** 4th Annual Internal Medicine Research Day, University of Kansas Medical Center (2023).
- **Second-Best Poster Winner:** Student Research Forum, University of Kansas Medical Center (2023).
- **Student Graduate Council Travel Award:** ASN Kidney Week, Orlando, FL (2022).

- **Elsevier Best Research Paper Award:** 43rd Annual Conference of Environmental Mutagen Society of India (2019).
- **Gold Medalist:** Best Research Paper, SRM University Research Day (2017).
- **Student Athlete:** Selected to represent the School Games Federation of India (SGFI) at 4 National Field Hockey Championship Tournaments (2006 to 2010).

Scientific Presentations

Invited Oral Presentations:

- **Chakraborty, A.** and Yu, A.S.L. Miniaturized CRISPR activation plasmid improves transgene delivery, PKD1 expression and induces signaling correction in polycystic kidney cells. Jared J. Grantham Symposium, Kansas City, KS (2025).
- **Chakraborty, A.** and Yu, A.S.L. Epigenetic reprogramming of PKD1 transcription: A proof of concept. Jared J. Grantham Symposium, Kansas City, KS (2023).
- **Chakraborty, A.** and Viswanathan, V. Ischemic acute kidney injury is characterized by heterogeneous methylomic changes in the renal cortex and medulla. 43rd Annual Conference for Environmental Mutagen Society of India, Vellore, TN, India (2019).
- **Chakraborty, A.,** Devaraju, P., Jayaraman, M. Analysis of selected upstream variants in TNF-alpha gene for their association with gastric mucosal inflammation in Indian Tamil patients with chronic Helicobacter pylori infection. SRM University Research Day, Chennai, TN, India (2017).

Poster Presentations:

- **Chakraborty, A.,** Wallace, D.P., Yu, A.S.L. Rationally engineered miniaturized plasmids improve transgene delivery and CRISPR-activation of PKD1 in primary kidney cells. ASN Kidney Week, Houston, TX (2025).
- **Chakraborty, A.** and Yu, A.S.L. Rationally Engineered Miniaturized Plasmids Improve Transgene Delivery and Expression in Hard-to-Transfect Primary Kidney Cells. Student Research Forum, KUMC, Kansas City, KS (2025).
- **Chakraborty, A.,** Ward, C.J., Yu, A.S.L. CRISPR activation of PKD1 in immortalized cell lines is limited by its heterochromatinized proximal promoter. ASN Kidney Week, Philadelphia, PA (2023).
- **Chakraborty, A.,** Ward, C.J., Yu, A.S.L. Heterochromatinized proximal promoter of PKD1 limits CRISPR activation of the gene in immortalized cell lines. PKD Connect Conference, Denver, CO (2023).
- **Chakraborty, A.,** Parnell, S.C., Zhang, Y., Daniel, E., Vrtek, K., Raman, A., Reif, G., Wallace, D.P. Early cyst formation leads to the development of an inflammatory microenvironment and tissue remodeling in polycystic kidney disease. ASN Kidney Week, Orlando, FL (2022).
- **Chakraborty, A.,** Ramar, M., Chellapan, D.R. Elucidation of receptor mediated action of eugenol from Cinnamomum tamala and Cinnamomum verum leaves against ileal spasms. 41st Indian Society of Human Genetics Annual Meeting, Chennai, TN, India (2016).
- **Chakraborty, A.** and Ramaswamy, R, M. Ex-vivo Characterization of Atheromatous Tissue Under SPION Based Magnetic Hyperthermia. International Conference on Emerging trends in Bioengineering, SASTRA Deemed University, TN, India (2013).

Academic Leadership

- **Planning Committee Member | Polycystic Kidney Disease-Renal Resource Consortium (PKD-RRC) Symposium (Apr 2023)**
 - Co-managed speaker scheduling, and technical execution for a national level scientific symposium, coordinating cross-institutional participation for renal researchers and trainees.

- **Vice President | Cell Biology & Physiology Society, University of Kansas Medical Center (Dec 2021 – Dec 2022)**
 - Led executive operations for departmental graduate trainees, organizing scientific seminars, professional development workshops, and networking events to enhance student career preparation and inter-departmental collaboration.
- **Interprofessional Committee Member | Student Governing Council, University of Kansas Medical Center (Aug 2022 – Aug 2024)**
 - Represented graduate student interests across medical and health professions schools, evaluating inter-professional grant proposals and advocating for campus-wide research resources.
- **Organizing Committee Member | 3-Minute Thesis, Student Research Forum, University of Kansas Medical Center (Jan 2022 – Dec 2024)**
 - Directed annual university-wide science communication competitions over a 3-year term, managing judge recruitment, workshop training for participants, and event execution to improve graduate student communication skills.
- **Chair | 8th Genetic Engineering Research Symposium, SRM Institute of Science and Technology (Apr 2016)**
 - Directed a multi-committee team to execute a national research symposium, managing budget allocation, abstract peer-review, and keynote speaker invitations for 100+ attendees.

Certificate and Professional Memberships

- **Professional Membership:** American Society of Nephrology (ASN) — Active Member (2018 – Present).
- **Professional Membership:** American Association for the Advancement of Science (AAAS; Mar 2024 – Feb 2025) & American Heart Association (AHA; Jan 2024 – Jun 2025).
- **Coursera / Johns Hopkins University:** Introduction to Genomic Technologies (2020).
<https://www.coursera.org/account/accomplishments/verify/66YYT3UT4KTQ>
- **Coursera / University of Melbourne:** Epigenetic Control of Gene Expression (2018).
<https://www.coursera.org/account/accomplishments/records/DX9LW662XBUX>